

Chapter 4

HTTP AND WEB SERVICES

4.1. HTTP, Web Servers and Web Access

4.1.1 HTTP

- HTTP (Hypertext Transfer Protocol) is a fundamental protocol used for data communication on the World Wide Web. It is the foundation of data communication on the internet, allowing for the transfer of various types of data, including text, images, videos, and more, between web servers and web clients (usually web browsers).
- Here are some key points about HTTP:
 - a. Request-Response Protocol: HTTP follows a client-server model, where a client (typically a web browser) makes requests to a server (where a website or web application is hosted), and the server responds with the requested data. The client initiates the communication by sending an HTTP request, and the server replies with an HTTP response.
 - b. Stateless: HTTP is a stateless protocol, which means that each request-response cycle is independent and does not retain any information about previous interactions. To maintain session state, web applications often use cookies or other mechanisms.

c. Methods: HTTP defines several request methods, such as GET (retrieve data), POST (submit data), PUT (update data), DELETE (remove data), and more. These methods allow clients to specify the type of operation they want to perform on a resource

d. URLs: Uniform Resource Locators (URLs) are used to specify the resource a client wants to interact with. A typical URL consists of a protocol (e.g., "http" or "https"), a domain name (or IP address) for the server, a port number (usually 80 for HTTP and 443 for HTTPS), and a path to the specific resource on the server.

e. Headers: HTTP requests and responses include headers that provide additional information about the data being transferred. Headers can include information like the content type, content length, caching instructions, and more.

f. Status Codes: HTTP responses include status codes that indicate the outcome of the request. For example, a 200 status code indicates a successful request, while a 404 status code indicates that the requested resource was not found.

g. Security: While HTTP is the basic protocol for web communication, it lacks encryption. For secure data transfer, HTTPS (HTTP Secure) is used, which incorporates SSL/TLS encryption to protect the data being exchanged between the client and server.

HTTP is a Stateless Protocol

- HTTP is indeed a stateless protocol. Each request from a client to a server is independent and does not carry any information about previous requests. The server does not maintain any knowledge of the client's state between requests. This statelessness simplifies the implementation of web servers and allows them to handle each request in isolation.
- To manage state and maintain user sessions, web applications often use techniques such as cookies or sessions. Cookies are small pieces of data stored on the client's side, which can be sent with subsequent requests to the same server. Sessions involve storing user-specific information on the server side and associating it with a session identifier sent to the client.
- The statelessness of HTTP contributes to its simplicity and scalability but requires additional mechanisms for handling user sessions and maintaining application state.

How HTTP Works?

- Client initiates TCP connection (creates socket) to server, typically port 80
- Server accepts TCP connection from client
- HTTP messages (application layer protocol message) exchanged between browser (HTTP client) and Web server (HTTP server)
- TCP connection closed

HTTP Header

Header appears in request and response to provide more information about the request or response. The types of header are:

- General Header
- Request Header
- Response Header
- Entity Header (optional)

1. General Header

Can be present in request or response

- Cache-Control: control resource caching
- Connection: specify connection options
- Date: time of request/response
- Transfer-Encoding: indicate message has been transformed ("chunked" is only option)
- Via: show hosts through which request or response has passed

2. Request Header

Meaningful only in requests. Important headers are:

- Accept: client tells server what media types are acceptable (e.g., text/html)
- Host: specifies host and port number for URL
- If-Modified-Since: used when performing conditional GET
- User-Agent: reports client software name and version.

3. Response Header

- Accept-Ranges: indicates server is capable of responding to range requests (e.g., byte range requests)
- Age: typically, the age of a cached object
- Location: used to indicate re-direction destination
- Server: identifies the server software that generated the response
- WWW-Authenticate: authentication challenge sent to client with "401 Unauthorized" response.

4. Entity Header

Carry meta information about the requested resource:

- Content-Base: defines base URL for relative URL in this document
- Content-Encoding: indicates encoding of entity (e.g., gzip or compress)
- Content-Length: length of entity in bytes
- Content-Type: media type of entity (e.g., text/html, image/gif etc)
- Expires: date/time at which entity expires (is no longer valid)
- Last-Modified: when entity modified on the origin server

Persistent and Non-Persistent HTTP Connection

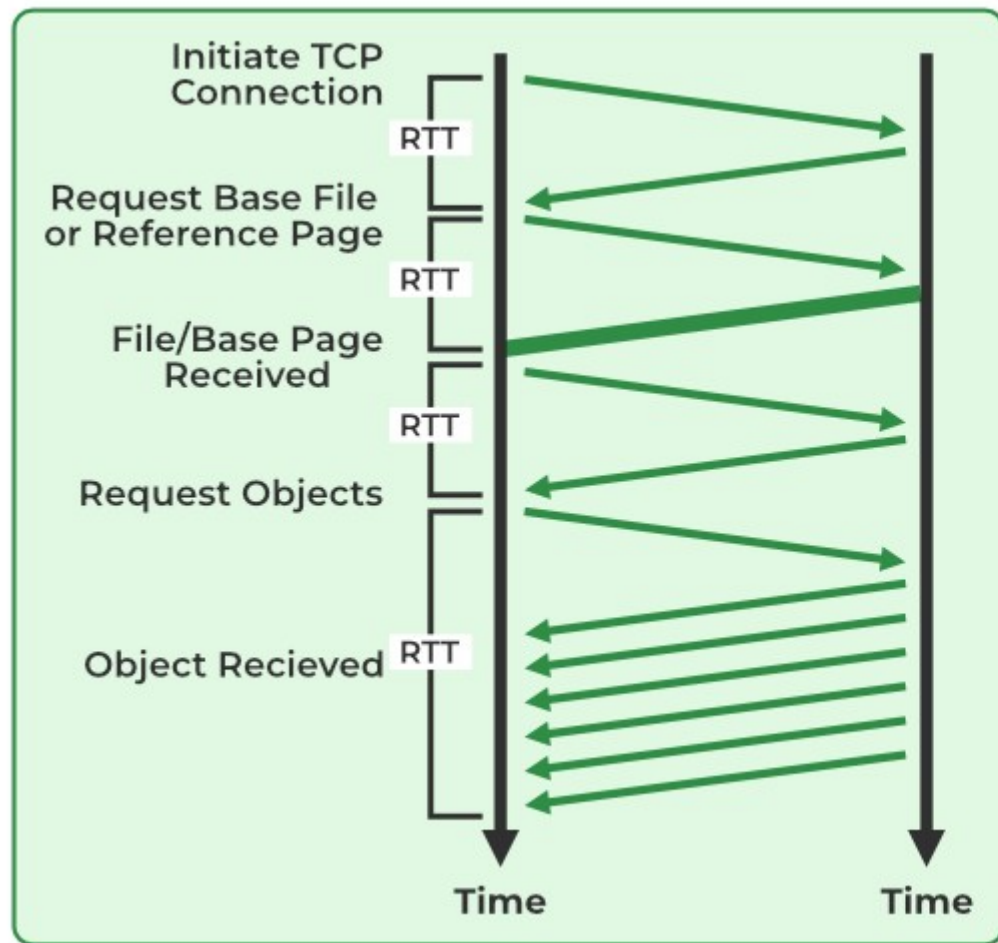
1. Non-Persistent Connection

- A non-persistent connection in the context of HTTP refers to a connection between a client (such as a web browser) and a server that is opened and closed for each individual request-response cycle. In other words, for each resource (like an HTML page, image, or script) that the client requests from the server, a new connection is established, and after the response is received, the connection is immediately closed.
- Key characteristics of non-persistent connections include:
 - a. Connection Overhead: Opening and closing connections for each request can introduce additional overhead. Establishing a new connection requires time and resources, and this process can impact the overall performance of the communication.
 - b. Limited Resource Reuse: Since connections are short-lived, resources such as TCP connections are not reused for multiple requests. This contrasts with persistent connections, where a single connection can be reused for multiple requests.
 - c. Simple Implementation: Non-persistent connections are simpler to implement, as the server doesn't need to keep track of the state between multiple requests from the same client. Each request is treated independently.
 - d. Default Behavior of HTTP/1.0: The original version of the HTTP protocol (HTTP/1.0) primarily used non-persistent connections by default. Each request created a new connection, and after the response, the connection was closed.

- Non-persistent connection HTTP is used in fetching those objects which are not needed that frequently.
- Non-persistent connection HTTP requires 2 RTT (Round Trip Time) for each object that is to be transmitted (1 RTT to open the connection and 1 RTT for transmission of data).
- While non-persistent connections are simpler, they can lead to higher latency and increased overhead due to the frequent opening and closing of connections. To address these issues, modern web browsers and servers often use techniques like connection keep-alive or persistent connections.
- Non-persistent connection is of two types:
 - a. With parallel connection
 - b. Without parallel connection

a. With parallel connection

- Parallel connections involve sending multiple requests simultaneously or in parallel over separate connections.
- This can be achieved by opening multiple connections to a server and sending requests concurrently.
- Parallel connections can be used to improve the overall performance and reduce latency, especially when a webpage requires multiple resources to load (e.g., images, scripts, stylesheets).
- Combining non-persistent connections with parallel connections means that, while each individual connection for a specific resource is short-lived, a client may open several connections concurrently to request multiple resources simultaneously.
- Non-persistent HTTP needs to maintain an extra overhead to open a TCP connection each time some data needs to be transmitted over it. This makes the process of exchange of HTTP requests and responses between a client and a server slower. To make this process faster, sometimes, browsers open parallel connections in which, after the opening of a TCP connection, multiple HTTP requests are sent from client to server parallelly. This saves the time of opening a new HTTP (TCP) connection for each HTTP request to the transmission of the object on the connection.



Non-Persistent & Parallel Connections

b. Without parallel connection

- Resources are fetched sequentially, one after the other.
- A new connection is opened for each resource, and the next resource is requested only after the previous connection is closed.
- In this setup, there is no simultaneous or parallel fetching of multiple resources. Each resource is fetched one at a time, and each connection is opened and closed independently for each resource.
- While this approach simplifies the implementation and may be suitable for simpler scenarios, it can result in higher latency and increased overhead due to the repeated process of establishing and closing connections for each resource. It may lead to a slower user experience, especially in situations where a webpage requires multiple resources to load.

Advantages of non-persistent connection:

- There is no wastage of resources since the connection is opened only when some data needs to be sent over it.
- It is more secure than persistent HTTP.

Disadvantages of non-persistent connection:

- It needs to maintain an extra overhead to open a TCP connection each time some data needs to be transmitted over it.
- It has a slow start because of the opening of a TCP connection on every data transmission.

2. Persistent Connection

- A persistent connection in the context of HTTP refers to a single network connection that is kept open between a client (such as a web browser) and a server across multiple HTTP request-response cycles. Unlike non-persistent connections, where a new connection is established and closed for each individual request, a persistent connection allows multiple requests and responses to be sent and received over the same connection.
- Key features of persistent connections include:
 - a. Connection Reuse: The same TCP (Transmission Control Protocol) connection is reused for multiple HTTP requests and their corresponding responses. This helps reduce the overhead associated with opening and closing connections for each request.
 - b. Lower Latency: Persistent connections can contribute to lower latency because the connection is already established, and subsequent requests can be sent without the delay of opening a new connection for each resource.
 - c. Improved Efficiency: Keeping the connection open allows for the efficient exchange of multiple resources, such as images, scripts, and stylesheets, without the need to establish a new connection for each resource.

d. Connection Keep-Alive: The term "Connection: keep-alive" is often used to describe persistent connections in the HTTP header. This header indicates that the connection should be kept open for multiple requests.

e. HTTP/1.1 Default Behavior: HTTP/1.1, the current version of the HTTP protocol, defaults to using persistent connections. In HTTP/1.0, the default behavior was non-persistent connections, and the "Connection: keep-alive" header was introduced to indicate a persistent connection.

- Persistent connections contribute to the optimization of web performance by reducing the time and resources required to establish and close connections for each individual request.
- Persistent HTTP does not require 2 RTT (round trip time) for each object that is to be transmitted. After a successful opening of the TCP connection (opening of TCP connection is done by 3-way handshaking which takes 1 RTT), each object will require only 1 RTT to be transmitted.
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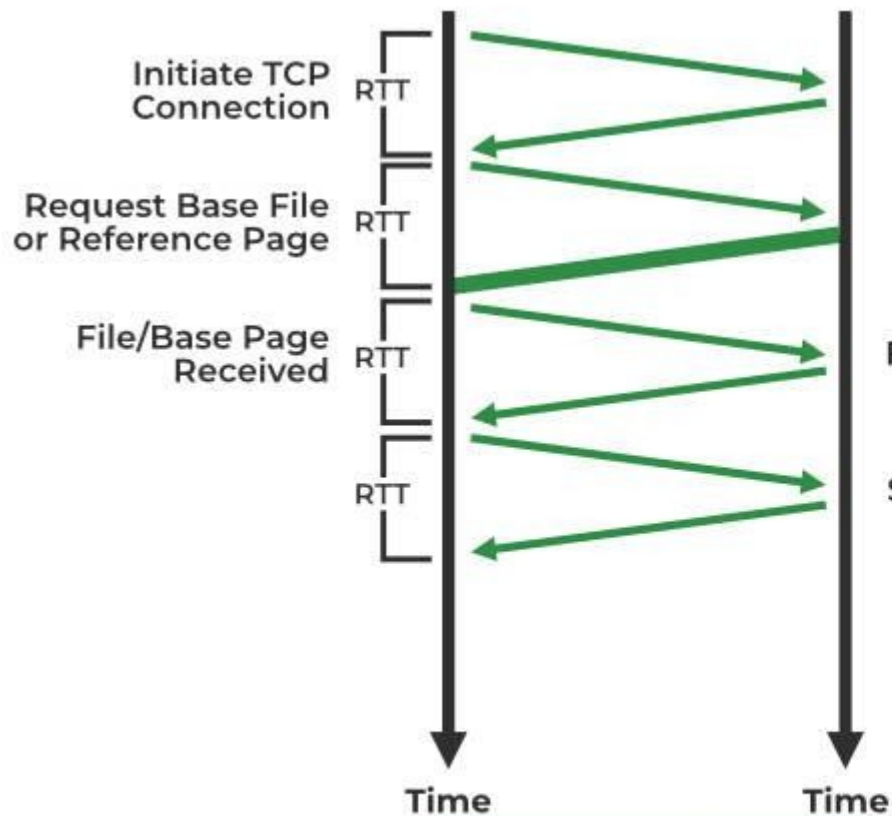
Persistent connection is of two types:

a. Non pipelined

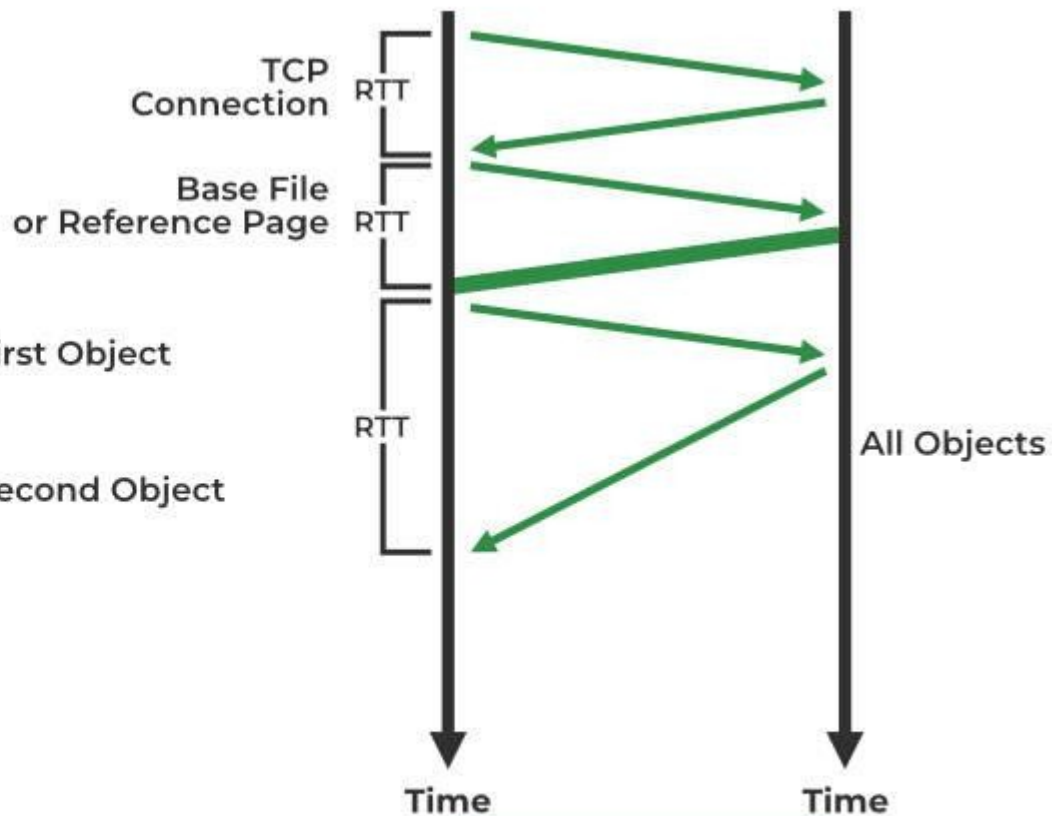
- In a non-pipelined HTTP connection, the client can send a request to the server only if the previous request that was sent by the client has been acknowledged by the server.

b. Pipelined

- In a pipelined HTTP connection, the client can send a request to the server even if the previous request that was sent by the client has not been acknowledged by the server.



**Persistent Without
Pipelining**



**Persistent With
Pipelining**

Advantages of persistent connection:

- It saves CPU resources and time since the opening of the connection takes place only once.
- It gives a fast start to send any object from a source to a destination.
- There is less network congestion and latency on subsequent HTTP requests compared to non-persistent HTTP.

Disadvantages of persistent connection:

- Persistent HTTP results in wastage of network resources since the connection remains open even when there is no transmission of data.
- It is less secure than non-persistent HTTP.

HTTP Request Methods

- HTTP (Hypertext Transfer Protocol) defines several request methods, also known as HTTP verbs, that indicate the desired action to be performed on a resource. Each request method has a specific purpose, and the server responds accordingly. Here are some commonly used HTTP request methods:
 - a. GET:
 - The GET method is used to request data from a specified resource.
 - It is a safe and idempotent method, meaning it should not have side effects, and multiple identical requests should produce the same result as a single request.
 - b. POST:
 - The POST method is used to submit data to be processed to a specified resource.
 - It is often used when uploading a file, submitting a form, or when the request may cause a change in the server's state.

c. PUT:

- The PUT method is used to update a resource or create a new resource if it does not exist.
- It is idempotent, meaning that making multiple identical requests has the same effect as a single request.

d. DELETE:

- The DELETE method is used to request the removal of a resource.
- It is idempotent, meaning that making multiple identical requests has the same effect as a single request.

e. PATCH:

- The PATCH method is used to apply partial modifications to a resource.
- It is typically used when updating a resource partially rather than replacing the entire resource.

f. HEAD:

- The HEAD method is similar to GET but is used to request the headers of a resource, not the actual data.
- It is often used to check the status of a resource without downloading the entire content.

g. OPTIONS:

- The OPTIONS method is used to describe the communication options for the target resource.
- It is often used to check the allowed methods on a resource or to discover server capabilities.

h. TRACE:

- The TRACE method is used to perform a message loop-back test along the path to the target resource.
- It is mainly used for diagnostic purposes and is less commonly used in practice.

i. CONNECT:

- The CONNECT method is used to establish a network connection with a resource, typically for the purpose of SSL/TLS tunneling.
- It is used in conjunction with the HTTP CONNECT method to establish a secure connection through an HTTP proxy.

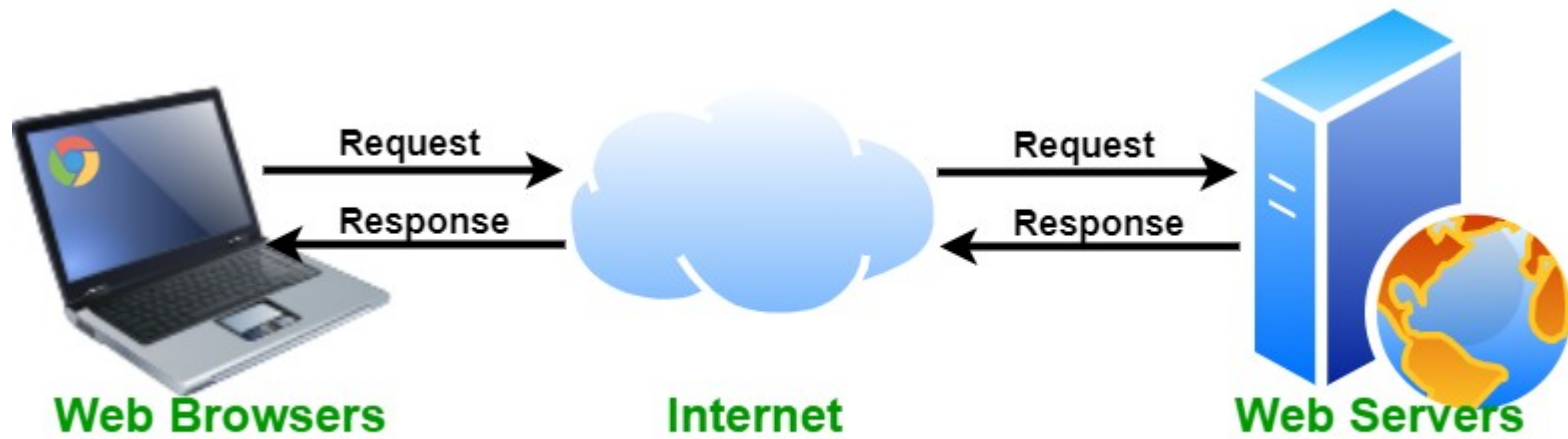
These HTTP request methods define the actions that clients (such as web browsers) want to perform on resources hosted on servers. The choice of method depends on the intended operation, and each method has its own set of rules and semantics.

4.1.2 Web Servers

- Web servers are software applications or programs that handle incoming requests from clients (such as web browsers) and serve the requested web pages or resources. They play a fundamental role in the architecture of the World Wide Web by facilitating the delivery of content to users.
- A Web server is a program that uses HTTP (Hypertext Transfer Protocol) to serve the files that form Web pages to users in response to their requests, which are forwarded by their computers' HTTP clients.
- A Web server consists of a physical server, operating system (OS) and software used to facilitate HTTP communication.
- Web server runs a website by returning HTML files over an HTTP connection. Web server is any Internet server that responds to HTTP requests to deliver content and services.
- Depending on context, the term can refer to the hardware or Web server software on the server.
- In terms of software, there have been literally hundreds of Web servers over the years, but Apache and Microsoft's IIS have emerged as two of the most popular systems.

How Web Servers Work?

- The browser breaks the URL into three parts: The protocol ("http"), the server name ("www.website.com"), the file name ("webpage.html").
- Obtaining the IP Address from domain name: The browser communicates with a name server (DNS), which translates the server name, into an IP address.
- Browser requests the full URL: The browser then forms a connection to the Web server at that IP address on port 80.
- Web server responds to request: Following the HTTP protocol, the browser sends a GET request to the server, asking for the file. The server sends the HTML text for the Web page to the browser.
- Browser displays the web page: The browser reads the tags and formats the page onto the screen.



4.1.3 Web Access

- Web Access Management (WAM) refers to the set of processes, technologies, and policies that are used to manage and control access to web resources within an organization. The primary goal of WAM is to ensure that users have the appropriate level of access to web applications, services, and data while maintaining security and compliance. WAM systems are commonly used in enterprise environments to govern user access to web-based resources.
- Web Access Management Architecture:
 - Plugins (Web Agent) are programs that are installed on every web/application server, register with those servers, and are called at every request for a web page. They intercept the request and communicate with an external policy server to make policy decisions. One of the benefits of a plugin (or agent) based architecture is that they can be highly customized for unique needs of a particular web server. One of the drawbacks is that a different plugin is required for every web server on every platform (and potentially for every version of every server). Further, as technology evolves, upgrades to agents must be distributed and compatible with evolving host software.

-Proxy-based architectures differ in that all web requests are routed through the proxy server to the back-end web/application servers. This can provide a more universal integration with web servers since the common standard protocol, HTTP, is used instead of vendor specific Application Programming Interfaces (APIs). One of the drawbacks is that additional hardware is usually required to run the proxy servers.

-Tokenization differs in that a user receives a token which can be used to directly access the back-end web/application servers. In this architecture, the authentication occurs through the web access management tool but all data flows around it. This removes the network bottlenecks caused by proxy-based architectures. One of the drawbacks is that the back-end web/application server must be able to accept the token or otherwise the web access management tool must be designed to use common standard protocols.

4.2 Universal Naming with URLs

4.2.1 Uniform Resource Identifier (URI)

- A Uniform Resource Identifier (URI) is a string of characters used to identify a resource on the internet. It can be further classified into two subsets: Uniform Resource Locators (URLs), which specify the location of a resource, and Uniform Resource Names (URNs), which serve as persistent identifiers regardless of the resource's location.
- URLs typically include information about the method used to access the resource (such as HTTP or FTP), the domain name or IP address where the resource is hosted, and the specific path or location of the resource on the server. For example, `https://www.example.com/page.html` is a URL that points to a web page named "page.html" on the "www.example.com" server using the HTTPS protocol. It is important to remember that URLs are not persistent. This means that if the resource's location changes, the URL also needs to change to point to the resource's new location.
- This type of URI does not state which protocol should be used to locate and access the resource; it simply labels the resource with a persistent, location-independent unique identifier. A URN will identify the resource throughout its lifecycle and will never change. URNs, on the other hand, are designed to be persistent identifiers for resources, meaning they stay the same even if the resource moves to a different location. An example of a URN might look like `urn:isbn:0451450523`, which could represent a specific book regardless of where it is hosted online.

4.2.2 Uniform Resource Locator (URL)

- A Uniform Resource Locator (URL) is the address of a resource on the Internet. A URL indicates the location of a resource as well as the protocol used to access it.
- URL is the unique address for a file that is accessible on the Internet. URL is the global address of documents and other resources on the World Wide Web.
- A common way to get to a website is to enter the URL of its home page file in our Web browser's address line. However, any file within that website can also be specified with a URL.
- A URL consists of the following components:
 - a. Protocol identifier: For the URL <http://example.com> , the protocol identifier is http, used to access the file resource.
 - b. Domain name or resource name: For the URL <http://example.com>, the resource name is example.com that identifies address of resource on the Internet.
 - c. Path name: a hierarchical description that specifies the location of a file in that computer.

The protocol identifier and the resource name are separated by a colon and two forward slashes.

The protocol identifier indicates the name of the protocol to be used to fetch the resource. E.g., HTTP, FTP etc.

The resource name is the complete address to the resource. The format of the resource name depends entirely on the protocol used, but for many protocols, including HTTP.

The resource name contains one or more of the following components:

- Host name: The name of the machine on which the resource lives.
- File name: The path name to the file on the machine.
- Port number: The port number to which to connect (typically optional).
- Reference: A reference to a named anchor within a resource that usually identifies a specific location within a file (optional).

4.3 WWW Technology: HTML, DHTML, WML, XML

4.3.1 WWW Technology

- The World Wide Web (WWW or simply the Web) is a system of interconnected hypertext documents accessed via the internet. It's based on the principles of hypertext transfer protocol (HTTP) and uses a standardized markup language called HTML (Hypertext Markup Language) to structure and link documents.
- Key technologies that form the foundation of the World Wide Web include:
 - i. HTML (Hypertext Markup Language): HTML is the standard markup language used to create documents on the web. It provides the structure and content of web pages using elements like headings, paragraphs, links, images, and more.
 - ii. HTTP (Hypertext Transfer Protocol): HTTP is the protocol used for transferring web pages and other resources across the internet. It defines how messages are formatted and transmitted, enabling communication between web servers and clients (browsers).
 - iii. URLs (Uniform Resource Locators): URLs are addresses used to access resources on the web. They specify the location of resources, such as web pages, images, videos, and more, allowing browsers to retrieve them from web servers.

iv. Web Browsers: Software applications like Chrome, Firefox, Safari, and others are used to access and display web content. Browsers interpret HTML documents, render them visually, and allow users to interact with web pages.

v. Web Servers: These are computers that store web content and deliver it to clients upon request. They use HTTP to respond to requests from browsers and serve web pages, images, videos, and other resources.

vi. CSS (Cascading Style Sheets): CSS is used to define the presentation and layout of HTML documents. It allows web developers to control the visual aspects of web pages, including colors, fonts, spacing, and more.

vii. JavaScript: JavaScript is a programming language used to add interactivity and dynamic behavior to web pages. It enables features like form validation, animations, and the manipulation of web page content

Functions of WWW:

- i. Linking: Most web pages contain hyperlinks to other related pages and perhaps to downloadable files, source documents, definitions and other web resources. In the underlying HTML, e.g.,
`<ahref="home.html">Example.orgHomepage</a>`.
- ii. Dynamic update of web pages: To make web pages more interactive, some web applications also use Stylesheet, JavaScript techniques, which is dynamically updates to viewer.
- iii. www prefix: When a user submits an incomplete domain name to a web browser in its address bar input field, some web browsers automatically try adding the prefix "www" to the beginning of it and possibly ".com", ".org" and ".net" at the end.

iv. Protocol identifier: The protocol identifiers `http://` and `https://` at the start of a web URI refer to Hypertext Transfer Protocol or HTTP Secure, respectively. They specify the communication protocol to use for the request and response. The HTTP protocol is fundamental to the operation of the World Wide Web, and the added encryption layer in HTTPS is essential when browsers send or retrieve confidential data, such as passwords or banking information. Web browsers usually automatically prepend `http://` to user entered URIs, if omitted.

A markup language is simply a set of rules that defines the layout, format, or structure of text within a document. After markup instructions are added to a document, the document must be read, or processed, by a program that knows how to interpret the markup elements.

4.3.2 HTML

- Hypertext Markup Language, commonly referred to as HTML, is the standard markup language used to create web pages.
- Along with CSS and JavaScript, HTML is a foundation of technology used by most websites to create visually engaging web pages, user interfaces for web applications, and user interfaces for many mobile applications.
- Web browsers can read HTML files and render them into visible or audible web pages.
- HTML describes the structure of a website semantically along with signs for presentation, making it a markup language, rather than a programming language.

- Here are some key aspects of HTML:

- a. Markup Language: HTML is a markup language, not a programming language. It uses tags to define the structure of content within a web page. Tags are enclosed in angle brackets (< >) and come in pairs - an opening tag and a closing tag - surrounding the content they affect.
- b. Structuring Content: HTML allows content to be organized into various elements such as headings, paragraphs, lists, images, links, tables, forms, and more. These elements provide the basic structure and semantics of a web page.
- c. Text and Multimedia: HTML handles text formatting, allowing the use of headings (h1 to h6), emphasis (using or tags), line breaks, and paragraphs. It also supports embedding multimedia content like images, audio, and videos.
- d. Hyperlinks: HTML enables the creation of hyperlinks, allowing users to navigate between different web pages. Links are created using the <a> tag, specifying the destination URL.
- e. Semantic Markup: HTML5 introduced a greater emphasis on semantic elements, such as <header>, <footer>, <nav>, <article>, <section>, and <aside>. These tags provide more meaningful structure and help search engines understand the content and its context better.
- f. Accessibility: Proper use of HTML ensures better accessibility for users with disabilities by providing semantic structure and support for assistive technologies like screen readers.
- g. Versioning: HTML has evolved over time, with different versions introducing new features and improvements. HTML5 is the latest major version, offering enhanced multimedia support, better semantics, and improved APIs.

4.3.3 DHTML

- Dynamic HTML is an umbrella term for a collection of technologies used together to create interactive and animated web sites by using a combination of a static markup language (such as HTML), a client-side scripting language (such as JavaScript), a presentation definition language (such as CSS), and the Document Object Model.
- DHTML allows authors to add effects to their pages by using scripting language. DHTML allows us to create web pages that are more animated and responsive to user interaction than HTML.
- Dynamic fonts are also a characteristic of DHTML. They allow web page designers to include font files, containing specific styles, sizes and colors with the page. Therefore, the font choice is not dependent on what the browser provides.
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DHTML is the combination of HTML, CSS and JavaScript.

- ◆ Animate text and images in their document, independently moving each element from any starting point to any ending point, following a predetermined path or one chosen by the user.
- Embed a ticker that automatically refreshes its content with the latest news, stock quotes, or other data.
- Use a form to capture user input, and then process, verify and respond to that data without having to send data back to the server.
- Include rollover buttons or drop-down menus.

4.3.4 WML

- WML stands for Wireless Markup Language. It's an XML-based markup language used for creating content for wireless devices, particularly for older mobile phones and devices with limited processing power and small screens. WML was a key technology used in the early days of mobile internet.
- It provides navigational support, data input, hyperlinks, text and image presentation, and forms, much like HTML.
- It allows you to present text portions of a web page on cell phones and PDAs (Personal Digital Assistants) by wireless access.

WML Features

- Small display
- Limited input capacity
- Narrow-band network connection
- Limited memory
- Limited CPU power
- It provides WML users with games, email services and instant messaging.

The following are some key features of WML as compared to HTML:

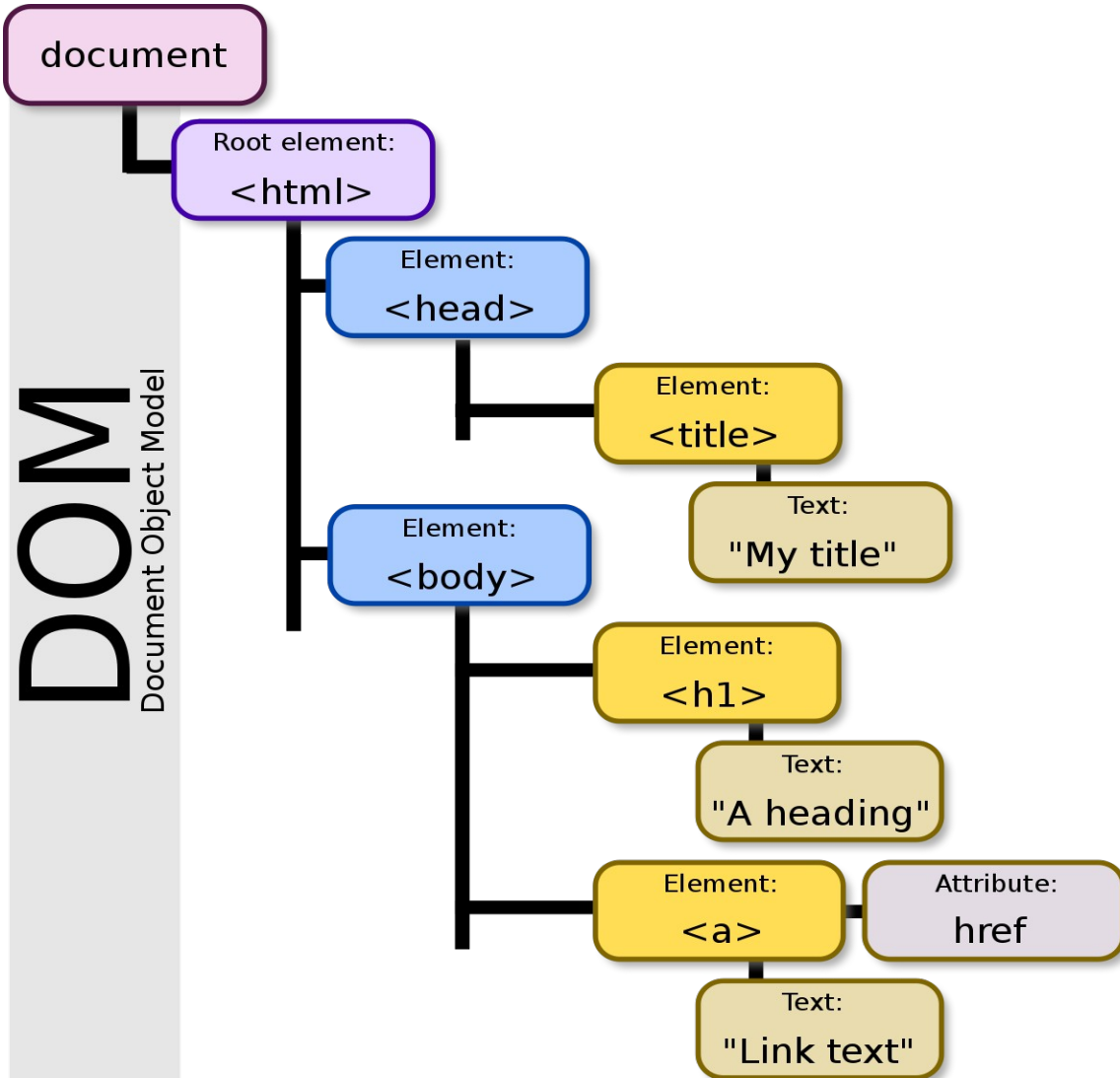
- WML is a markup language for small, wireless computing devices.
- In WML, variables can be defined that store data in string format. In HTML, variables cannot be stored.
- WML uses WML script for client-side scripting, which is stored in a separate file. HTML uses JavaScript.
- The supported image format for WML is WBMP. HTML supports JPEG, GIF and BMP.
- A micro-browser is used to run WML markup. A regular browser, such as Internet Explorer, Firefox or Chrome, is used to run HTML markup.
- WML follows XHTML specification and is therefore case sensitive. HTML is not case sensitive.
- WML has fewer tags compared to HTML.

4.3.5 XML

- XML, or eXtensible Markup Language, is a versatile markup language designed to store and transport data. It's similar to HTML in syntax but differs in its purpose and flexibility.
- It lets us define our own tags, the order in which they occur, and how they should be processed or displayed.
- XML is a markup language, user defined language that defines a set of rules for encoding documents in a format which is both human readable and machine-readable.
- The design goals of XML emphasize simplicity, generality and usability across the Internet. It is a textual data format with strong support via Unicode for different human languages.
- Although the design of XML focuses on documents, it is widely used for the representation of arbitrary data structures such as those used in web services.
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4.3.6 DOM

- The Document Object Model (DOM) is a programming interface for web documents. It represents the structure of a web page as a tree-like structure where each node represents a part of the document, such as elements, attributes, and text. The DOM provides a way for programming languages like JavaScript to interact with the structure, content, and style of web documents.
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Key aspects of the DOM:

- **Tree Structure:** The DOM organizes the elements of a web page in a hierarchical tree structure. Each node in the tree corresponds to an element, attribute, or text in the HTML document.
- **Nodes:** Nodes can be elements (like `<div>`, `<p>`, `<span>`), attributes (like `id="example"`), or text content. They can be accessed, modified, added, or removed using DOM methods and properties.
- **API for Interactivity:** The DOM provides methods and properties that allow developers to manipulate the content, structure, and style of a web page dynamically. It enables the creation of interactive web applications by allowing JavaScript to interact with and modify the document.
- **Platform Independence:** The DOM is platform-independent, meaning it provides a standardized way to access and manipulate documents regardless of the platform or programming language used.
- **Event Handling:** The DOM allows for event handling, enabling developers to define actions or functions that should be triggered in response to user interactions or certain events on the web page (like clicks, mouse movements, or key presses).
- **Document Manipulation:** Developers can dynamically create, modify, or delete elements, attributes, and content within a web page using DOM methods. This dynamic manipulation enables the creation of responsive and interactive web applications.

4.3.7 XHTML

- XHTML is a standard proposed by the W3C that adapts HTML into an extensible concept by using XML (Extensible Markup Language).
- XML defines data that can be shared on the web. It is extensible because anyone can invent a set of purposes such as describing the appearance of a web page. To enhance web pages, HTML was redesigned by using XML to form XHTML.
- XHTML is portable to enable small devices to support embedded programming. XHTML brings different programming practices. It has strict code rules such as symmetrical form, use lowercase, enclose elements with quotes and end tag with a forward slash at the end of the element and before the closing angle bracket.

- Difference between XML and HTML
- Difference between HTML and DHTML
- Differences between WML and HTML

4.4 Tools: WYSIWYG Authoring tools

- "What You See Is What You Get" (WYSIWYG) refers to an interface or editor that allows users to see a visual representation of what the final output will look like while they are editing or creating content. In other words, the displayed content in the editor closely resembles how it will appear when published or printed.
- Web authoring tools are used to create web content, and cover a wide range of software programs you can download to your computer or access online. Today's web authoring tools can provide the power to build an interactive, animated, state-of-the-art website suitable for anything from personal web page to corporate business site.
- WYSIWYG mimics how something will appear, giving the user the opportunity to return to the editing state for any changes or modifications that might be required before the work is turned into a Web page, printed document or slide presentation.
- WYSIWYG is especially popular when it comes to web-publishing.
- The commonly used three web authoring tools are:
 - NetObjects Fusion
 - Microsoft Frontpage
 - Macromedia Dreamweaver

- WYSIWYG editors are commonly used in various applications, including:
 - a. Word Processors: Programs like Microsoft Word, Google Docs, or Apple Pages often use WYSIWYG interfaces, where users can format text, insert images, create tables, and adjust layout elements directly in a way that closely resembles the final printed or displayed document.
 - b. Website Builders: Some website creation tools, like WordPress, Wix, or Squarespace, offer WYSIWYG editors that allow users to design web pages visually without needing to know HTML or CSS. Users can drag and drop elements onto a canvas and see the layout and design in real time.
 - c. Email Editors: Email marketing platforms often provide WYSIWYG editors for creating email campaigns. Users can design the email layout, add text, images, and formatting, seeing a representation of the email's appearance as they work.
 - d. Graphics and Design Software: Applications like Adobe Photoshop, Illustrator, or InDesign provide WYSIWYG interfaces for designing graphics, illustrations, and layouts.

- WYSIWYG editors aim to simplify the creation process by providing a visual representation that closely mirrors the final output. This visual feedback helps users make design decisions and formatting choices more intuitively, as they can see the immediate impact of their changes. However, sometimes there might be slight discrepancies between what is shown in the editor and the final output due to differences in display settings, browsers, or other factors.

4.4.1 Types of Web Authoring Tools

- Web authoring tools encompass a wide range of software designed to assist in creating, designing, and managing content for the web. These tools vary in complexity, features, and purpose. Here are different types of web authoring tools:
 - a. Text Editors: Basic text editors like Notepad (Windows), TextEdit (Mac), or more advanced ones like Sublime Text, Atom, or Visual Studio Code. While not specifically designed for web development, they're commonly used for writing HTML, CSS, JavaScript, and other web languages.
 - b. WYSIWYG Editors: What You See Is What You Get editors like Adobe Dreamweaver, Microsoft Expression Web, or BlueGriffon. These offer visual interfaces allowing users to design websites without extensive coding knowledge, using drag-and-drop features and visual layout tools.
 - c. Content Management Systems (CMS): Platforms like WordPress, Joomla, Drupal, or Squarespace provide tools for creating and managing websites. They offer user-friendly interfaces for content creation, publishing, and site management without requiring extensive technical skills

d. Graphic Design Tools: Software such as Adobe Photoshop, Illustrator, or Sketch, used for creating graphics, logos, icons, and other visual elements that can be integrated into web pages.

e. Frameworks and Libraries: Tools like Bootstrap, Foundation, or MaterializeCSS provide pre-designed UI components, styles, and layouts to streamline web development by offering ready-to-use templates and elements.

f. Integrated Development Environments (IDEs): Advanced tools like WebStorm, Visual Studio, or Eclipse specifically designed for web development. They offer code editors, debugging tools, version control integration, and more to facilitate efficient coding.

g. Browser Developer Tools: Built-in tools in web browsers like Google Chrome's Developer Tools or Firefox's Developer Edition, used for inspecting and debugging web pages. They allow developers to modify HTML, CSS, and JavaScript in real time for testing and debugging purposes.

h. Online Website Builders: Platforms like Wix, Weebly, or Shopify offer online tools that enable users to create websites through a visual interface without requiring coding skills.

4.5. Helper Applications: CGI, PERL, JAVA, JAVASCRIPT, PHP, ASP, .NET Applications

4.5.1 CGI (Common Gateway Interface)

- The Common Gateway Interface (CGI) is a standard protocol that defines how web servers communicate with external programs or scripts to generate dynamic content or perform other server-side tasks. It's one of the oldest methods for creating dynamic web content and interacting with web servers.
- Key points about CGI:
 - a. Dynamic Content Generation: CGI enables web servers to execute programs or scripts in response to client requests. These programs generate dynamic content, such as web pages, based on user input, database queries, or other computations.
 - b. Server-Client Interaction: When a server receives a request for a CGI script (often through an HTTP request), it launches the script as a separate process. The script processes the request, generates the content, and sends it back to the server, which then delivers it to the client.
 - c. Programming Language Independence: CGI scripts can be written in various programming languages like Perl, Python, PHP, C, and others. As long as the server supports CGI, it can execute scripts written in any compatible language.

- d. Environment Variables: CGI scripts can access environmental variables containing information about the request, such as HTTP headers, request method, query parameters, and client IP address.
- e. File-Based Interaction: CGI typically operates by having the web server pass data between the client's request and the CGI script via standard input/output and environment variables. This means data is often passed as text files or streams between the server and the script.
- f. Security Concerns: Poorly designed or vulnerable CGI scripts can pose security risks, such as allowing unauthorized access, injection attacks, or performance issues if not optimized.

To understand the concept of CGI, let's see what happens when we click a hyperlink to browse a particular web page or URL.

- Your browser contacts the HTTP web server and demand for the URL i.e., filename.
- Web server will parse the URL and will look for the filename, if it finds that file then sends back to the browser, otherwise sends an error message indicating that you have requested a wrong file.
- Web browser takes response from web server and displays either the received file or error message.

4.5.2 PERL

- PERL is a high-level, general-purpose, interpreted, dynamic programming languages. The languages in this family include Perl 5 and Perl 6.
- The Perl languages borrow features from other programming languages including C, shell script (sh), AWK, and sed.
- In addition to CGI, Perl 5 is used for system administration, network programming, finance, bioinformatics, and other applications, such as for GUIs.
- It has become the language choice for WWW development, text processing, Internet services, mail filtering, graphical programming and every other task requiring portable and easily- developed solutions.

How Perl Works?

- When the Perl compiler is fed a Perl program, the first task it performs is lexical analysis: breaking down the program into its basic syntactic elements (often called tokens).
- If the program is: `print "Hello, world! \n";` the lexical analyzer breaks it down into three tokens: `print`, `"Hello, world! \n"`, and the final semicolon.
- The token sequence is then parsed, fixing the relationship between the tokens. In Perl, the boundary between lexical analysis and parsing is blurred more than in other languages.

- Once a program has been parsed and understood, it is compiled into a tree of opcodes representing low-level operations, and finally that tree of operations is executed unless you invoked Perl with the -c ("check syntax") switch, which exits upon completing the compilation phase.
- It is during compilation, not execution, that BEGIN blocks, CHECK blocks, and use statements are executed.

Typical uses of Perl:

I. Text processing

- The main use of Perl is text processing. It is used to manipulate textual data, reports, emails, news articles, log files, or just about any kind of text.

ii. System administration tasks

- It is useful for tying together lots of smaller scripts, working with the file systems, networking, and so on.

iii. CGI and web programming

- HTML is just text with built-in formatting, Perl can be used to process and generate HTML. Perl was de facto language for web development, and is still heavily used today.
- There are many freely available tools and scripts to assist with web development in Perl.

iv. Database Interaction

- Perl's DBI module makes interacting with all kinds of databases – ex: oracle, it is easy and portable.
- It is increasingly being used to write large database applications, especially those which provide a database backend to a website.

v. Other Internet programming

- Perl modules are available for just about every kind of internet programming, from mail and news clients, interfaces to Internet Relay Chats (IRC), right down to lower-level socket programming.

4.5.3 JAVA

- Java is a general-purpose computer programming language that is concurrent, class-based, object-oriented and also platform independent.
- It is intended to let application developers "Write Once, Run Anywhere" (WORA), meaning that compiled Java code can run on all platforms that support Java without the need for recompilation.

Key features and aspects of Java include:

- **Platform Independence:** Java programs are compiled into an intermediate form called bytecode, which can run on any system with the Java Virtual Machine (JVM). This "write once, run anywhere" capability contributes to its platform independence.
- **Object-Oriented:** Java is fundamentally object-oriented, emphasizing the use of objects and classes. It supports concepts like encapsulation, inheritance, and polymorphism, fostering modular and reusable code.
- **Rich Standard Library:** Java provides a vast standard library known as the Java API (Application Programming Interface), offering a wide range of prebuilt classes and packages for various functionalities, such as networking, I/O, utilities, and data structures.
- **Security:** Java incorporates strong security features, including a sandbox environment for running untrusted code and built-in security mechanisms to protect against vulnerabilities like buffer overflows and unauthorized access.

- **Multithreading:** Java supports multithreading, allowing concurrent execution of multiple tasks within a program. This capability is useful for developing applications that benefit from parallelism and handling multiple operations simultaneously.
- **Garbage Collection:** Java manages memory automatically through its garbage collection mechanism. This feature frees developers from manual memory management, reducing the risk of memory leaks and making the language more reliable.
- **Widespread Use:** Java is used in a variety of domains, including enterprise software development, web applications, mobile app development (Android), scientific computing, financial services, and more.
- **Frameworks and Ecosystem:** Java has a rich ecosystem with numerous frameworks and tools like Spring, Hibernate, Maven, and others, which simplify development tasks, facilitate web application development, and provide additional functionalities.

4.5.4 JavaScript

- JavaScript is a versatile, high-level programming language primarily known for its use in web development. Developed in the mid-1990s by Netscape, JavaScript has evolved into a powerful language used for both front-end and back-end development, as well as for creating desktop and mobile applications.
- Key aspects and features of JavaScript include:
 - i. Client-Side Scripting: JavaScript is primarily used in web browsers to add interactivity, manipulate the DOM (Document Object Model), and dynamically change content on web pages. It allows for interactive user experiences by responding to user actions such as clicks, form submissions, and more.
 - ii. Versatility: With the introduction of Node.js, JavaScript expanded its capabilities to server-side programming. Node.js allows developers to use JavaScript to build server-side applications, enabling full-stack development using a single language.

- **Asynchronous Programming:** JavaScript utilizes asynchronous programming, leveraging callbacks, promises, and `async/await` syntax to handle tasks without blocking the execution of other code. This is particularly useful for handling I/O operations and improving performance.
- **Dynamic Typing:** JavaScript is dynamically typed, meaning variable types are determined at runtime. This flexibility simplifies coding but can also lead to errors if not handled carefully.
- **Extensive Libraries and Frameworks:** The JavaScript ecosystem boasts numerous libraries and frameworks such as React, Angular, Vue.js (for front-end development), Express.js, and NestJS (for back-end development). These tools simplify development tasks, provide structure, and enhance productivity.
- **JSON (JavaScript Object Notation):** JavaScript's syntax is closely related to JSON, making it easy to work with JSON-formatted data, which is commonly used for data interchange between web servers and clients.
- **Browser Compatibility:** JavaScript is supported by all modern web browsers, allowing developers to create cross-browser applications and websites.
- **Community and Support:** JavaScript has a large and active developer community contributing to its growth, with a wealth of resources, tutorials, and open-source projects available.

Advantages of JavaScript:

- Less server interaction: We can validate user input before sending the page off to the server. This saves server traffic, which means less load on our server.
- Immediate feedback to the visitors: We don't have to wait for a page reload to see if we have forgotten to enter something.
- Increased interactivity: We can create interfaces that react when the user hovers over them with a mouse or activates them via the keyboard.
- Richer interfaces: We can use JavaScript to include such items as drag-and-drop components and sliders to give a rich interface to your site visitors.

Limitations of JavaScript:

- Client-side JavaScript does not allow the reading or writing of files. This has been kept for security reason.
- JavaScript cannot be used for networking applications because there is no such support available.
- JavaScript doesn't have any multi-threading or multiprocessor capabilities.

4.5.5 PHP

- PHP, which stands for Hypertext Preprocessor (originally Personal Home Page), is a server-side scripting language designed primarily for web development. It's a powerful and widely used language for creating dynamic web pages, handling form data, interacting with databases, and building web applications.
- PHP supports a large number of major protocols such as POP3, IMAP, and LDAP. PHP4 added support for Java and distributed object architectures (COM and CORBA), making n-tier development a possibility for the first time.
-

Common uses of PHP:

- PHP performs system functions, i.e., from files on a system, it can create, open, read, write, and close them.
- PHP can handle forms, i.e., gather data from files, save data to a file, through email you can send data, return data to the user.
- You add, delete, modify elements within your database through PHP.
- You can access cookies variables and set cookies.
- Using PHP, you can restrict users to access some pages of your website.
- It can encrypt data.

Characteristics of PHP:

- Five important characteristics make PHP's practical nature possible.
- They are:
 - Simplicity
 - Efficiency
 - Security
 - Flexibility
 - Familiarity

How does PHP works?

- PHP is a server-side scripting language which means code will be executed in server and then server sends only plain HTML code to browser. Browser can understand only HTML, CSS, JAVASCRIPT which are Client-Side-Scripting Languages.
- Here is the Step by Step Process
 - ✦ Client will send a HTTP/HTTPS request to Apache server.
 - ✦ PHP engine executes that commands.
 - ✦ And finally, server sends HTML output to client.
 - ✦ In client system, HTML will be executed by the client Application (e.g., browser) and shows output.

4.5.6 ASP

- An Active Server Page (ASP) is an HTML page that includes one or more scripts (small embedded programs) that are processed on a Microsoft web server before the page is sent to the user.
- An ASP is somewhat similar to a server-side or a common gateway interface (CGI) application in which all involve programs that run on the server, usually tailoring a page for the user.
- ASP.NET is a unified web development model integrated with .NET framework, designed to provide services to create dynamic web applications and web services.
- It is built on the Common Language Runtime (CLR) of the .NET framework and includes those benefits like multi-language interoperability, type safety, garbage collection and inheritance.

Features of ASP

- It has limited OOPs support and not having built in support for XML.
- Very less development and debugging tool available, meaning that is difficult to debug the code.
- We can only do scripting using visual basic scripting and Java scripting.
- Error handling is very poor.
- It has no high-level programming structure. Mixed of html and server-side scripting.
- We must be entering first line as: `<%LANGUAGE="VBSCRIPT" CODEPAGE="960"%>`
- It has no in-built validation control, meaning that validating page is difficult for developers.
- In the classic ASP, if we need to update code on the existing page then it is mandatory to restart the server to get reflect.

4.6 Introduction to AJAX (PROGRAMMING)

- AJAX stands for Asynchronous JavaScript and XML. It's a set of web development techniques used to create asynchronous web applications. AJAX allows web pages to send and receive data from a server without having to reload the entire page. This technique is commonly used to build more dynamic and interactive web applications by making background requests to the server, retrieving data, and updating specific parts of the web page without refreshing the whole thing. It's often implemented using a combination of JavaScript, XML, JSON, HTML, and CSS. However, nowadays, JSON is more commonly used than XML for data interchange due to its simplicity and ease of use.

4.6.1 How does AJAX work?

- AJAX calls are asynchronous, meaning they're made behind the scenes, independently from each other and the site as a whole. When a browser makes an AJAX call to the server, it isn't stuck waiting for a response, halting all of the site's functionality. Instead, the web service on the server will send the data back to the browser once the task is completed, where client-side scripts will process the response and deliver it to the user.
- AJAX is implemented using the following technologies:
 - JavaScript: client-side scripting language that ties all of the technologies together.
 - HTML/XHTML/CSS: use for the presentation of the data
 - Document Object Model (DOM): used to access and manipulate structured documents
 - XML/JSON: used to exchange data. JSON is now commonly used in place of XML format.
 - XMLHttpRequest: JS object that is responsible for asynchronous communication with the server.

4.6.2 Advantages of AJAX

- Improved user experience: Ajax makes it possible to create interactive applications that are fast and don't require reloading the whole pages.
- Reduced bandwidth Usage: With AJAX callbacks, the server won't get overloaded by requests, which would otherwise cause the app to slow down.
- Improved system performance: AJAX only retrieves the required data from the server. This greatly improves the system performance and response time.
- Promotes separation of data, business logic and presentation: Ajax calls usually retrieve data from the server and if necessary, business logic is applied. Data is displayed after these activities have successfully been completed.
-

- Better speed and performance
- More responsive, user-friendly websites and applications
- Ease of development, thanks to JSON, jQuery, and compilers that help to streamline complex JavaScript code
- Browser and platform-independent. Also, Microsoft offers the AJAX Control Toolkit integrated into Visual Studio on the ASP.NET platform
- Ideally implemented for updating small bits of information within a web application
- Allows text box autocomplete and predictive text

4.6.3 Disadvantages of AJAX

- Requires JavaScript: JavaScript is a client-side technology and we have no control over it. If the user disables JS on their web, then Ajax will not work.
- Web browser compatibility: Not all the web browser especially very old ones have support for all the technologies that AJAX uses.
- Hard/impossible to bookmark content: With Ajax content, this is impossible or at a minimum requests extra effort to implement.
- JavaScript content generally isn't SEO friendly: Developing SEO content with JS requires extra efforts.

4.7 Browser as a Rendering Engine

4.7.1 Browser

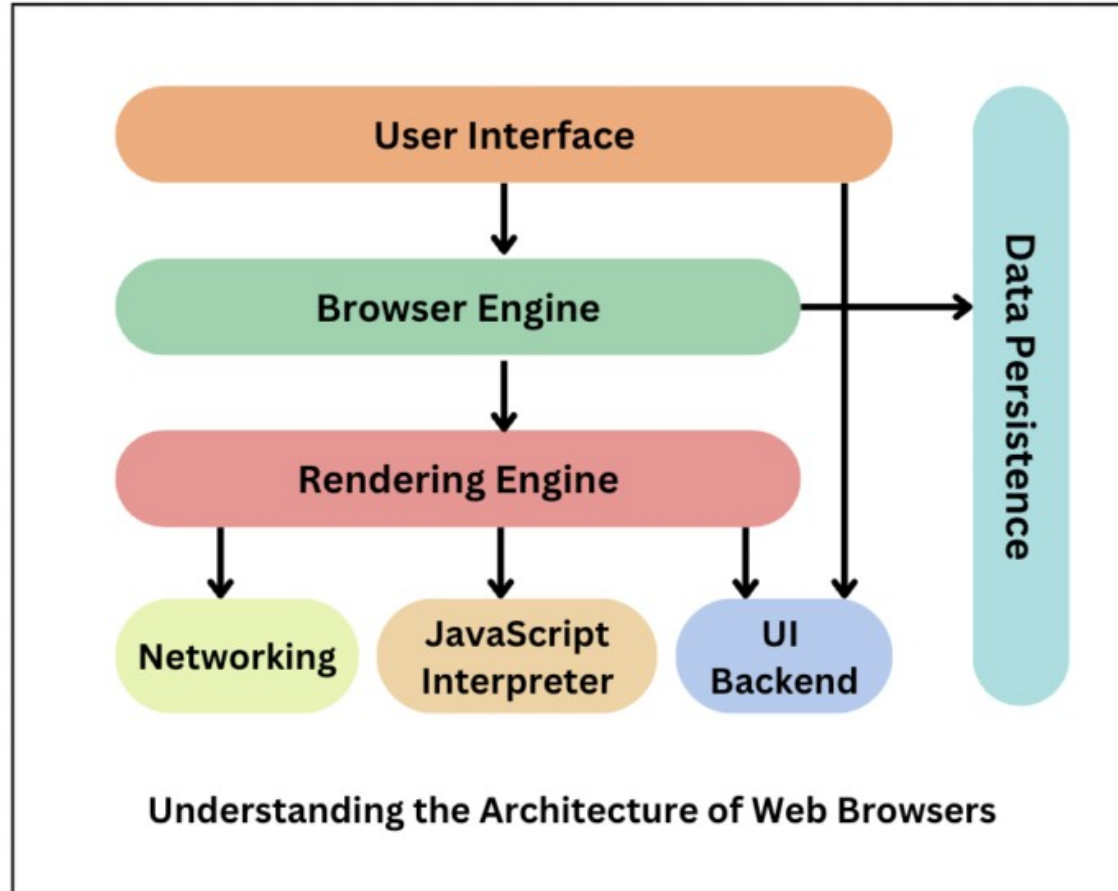
- A browser, short for web browser, is a software application used to access and display content on the World Wide Web. It's the primary tool people use to navigate the internet and view websites, web pages, images, videos, and other online content.
- Key functions of a browser include:
 - Fetching Web Pages: Browsers retrieve data from servers across the internet, including HTML documents, images, videos, stylesheets, and scripts, to display web pages.
 - Rendering: They interpret the HTML, CSS, and JavaScript code to render the web page layout and content as intended by the website developers.

-Navigation: Users can enter web addresses (URLs) into the browser's address bar to navigate to specific websites or use bookmarks, history, and tabs to move between different pages.

-Interactivity: Browsers enable user interaction with web pages through clicking links, filling out forms, submitting data, and executing scripts.

-Support for Standards and Technologies: They support various web standards like HTML, CSS, JavaScript, and also implement technologies for security (like HTTPS), multimedia playback, and more.

4.7.2 Components of Browser



- User interface: User interface is the space where interaction between users and the browser occurs. This includes the address bar, back/forward button, bookmarking menu, etc.
- Browser engine: Browser engine is the piece of code that acts as a bridge between the user interface and the rendering engine. It is responsible for querying and manipulating the rendering engine according to the inputs from various user interfaces.
- Rendering engine: Rendering engine is responsible for rendering a specific web page requested by the user on their screen. By default, the rendering engine can display HTML and XML documents and images. It can display other types of data via plug-ins or extension; for example, displaying PDF documents using a PDF viewer plug-in. Every browser has its own unique rendering engine. Also, different versions of a browser might have different rendering engine.

- When a user request a particular document, the rendering engine performs the following steps:
 - Parses the requested HTML page to construct a DOM tree
 - Constructs a render tree. The render tree includes both styling and visual information that define the order in which the elements will be displayed
 - Render tree goes through the layout process, in which the node is assigned the exact coordinates so that every node appears at an accurate position on the screen.
 - Paints the render tree on the screen.

- Networking: This component is responsible for managing network calls using standard protocols like HTTP or FTP. For example, sending the http requests to the server.
- JavaScript interpreter: It is the component of the browser written to interpret the JavaScript code presented in a web page.
- UI backend: It is used for drawing basic widgets like combo boxes and windows. This backend exposes a generic interface that is not platform specific.
- Data persistence/storage: This is a persistence layer. It is small database created on the local drive of the computer where the browser is installed. This database stores various files like cache, cookies, etc.

4.9 Web Hosting

- Web hosting refers to the service that allows individuals or organizations to make their websites accessible via the World Wide Web. When you create a website, the files, data, images, and other content need to be stored on a server that is connected to the internet. This is where web hosting comes into play.

4.9.1 Types of Web Hosting

- There are 3 types of web hosting. They are:

I. Shared Web Hosting:

- In shared web hosting, multiple web site owners shared a single server.
- It provides cost effective hosting as the server cost is shared among many owners.
- The performance of the web site is affected by other web sites who share the server and its resources.

ii. Dedicated Web Hosting

- In dedicated web hosting, the web site owner has a single web server rented for a single site.
- The owner has full control over the server.
- It is very expensive to rent a dedicated server.
- It provides high performance of the web site to the web traffics.

iii. Virtual Web Hosting

- Virtual web hosting is the bridge between shared and dedicated web hosting.
- In virtual hosting, multiple web sites share the resources of a single web server.
- But each web site is partitioned off as if it is hosted in the dedicated web server.
- The web site owner will have more control over sub domains and other features.
- It is cheaper than dedicated hosting but expensive than shared hosting.
- It is perfect for the web sites with fair amount of web traffics.

4.9.2 Virtual Hosting

- Virtual hosting is a method of hosting multiple domain names on a server using a single IP address. This allows one server to share its resources, such as memory and process cycles, in order to use its resources more efficiently. The term virtual hosting is usually used in reference to web servers but the principles do carry over to other internet services.
- There are 3 types of virtual web hosting possible in Apache:
 - Port based
 - Name based
 - IP based

1. Port based Virtual Web Hosting

- The default port number for HTTP is 80. However, most web servers can be configured to operate on almost any port number, provided the port number is not in use by any other program on the server. There is the HTTP Secure special port 443 that needs special configuration. Port based web sites are explicitly bound to a unique port number and an IP address. In this case the IP address is used for hosting multiple web sites. The unique port number used for a common IP address distinguish an individual web site from other websites bound to the same IP address.

2. IP based Virtual Web Hosting

- When IP based virtual hosting is used, each site (either a DNS host name or a group of DNS host names that act the same) points to a unique IP address. The webserver is configured with multiple physical network interfaces, virtual network interfaces on the same physical interface or multiple IP addresses on one interface. As the term IP-based indicates, the server must have a different IP address/port combination for each IP-based virtual host. The downside of this approach is the server needs a different IP address for every web site. This increases administrative overhead (both assigning addresses to servers and justifying the use of those addresses to internet registries) and contributes to IPv4 address exhaustion.

3. Name based virtual Web hosting

- Name based virtual hosts are multiple host names for the same web server IP address. A technical prerequisite needed for name-based virtual hosts is a web browser with HTTP/1.1 support (commonplace today) to include the target hostname in the request. With the name based virtual hosting, you can host several domains/websites on a single machine with a single IP. All domains on that server will be sharing a single IP. It's easier to configure than IP based virtual hosting, you only need to configure DNS of the domain to map it with its correct IP address and then configure Apache to recognize it with the domain names.